Annotation Guidelines

ADV layout segmentation

This guide should be consulted and strictly followed for all annotations for the purpose of training data generation for automatic layout recognition in the ADV project. The following annotation software is recommended BoundingBoxEditor.

Basic rules

We use simple rectangular frames for our markings.

The rectangles are always aligned parallel to the bottom edge of the scan. If the scan is skewed, the rectangle does not rotate accordingly.

Nevertheless, a marking should always cover the entire feature to be annotated – as closely as possible, but as generously as necessary.

Only the header section and the answers are marked. The questions themselves or other pre-printed areas are ignored unless they are used to answer the questions. For example, underlining in the questions.

Bounding Box Editor

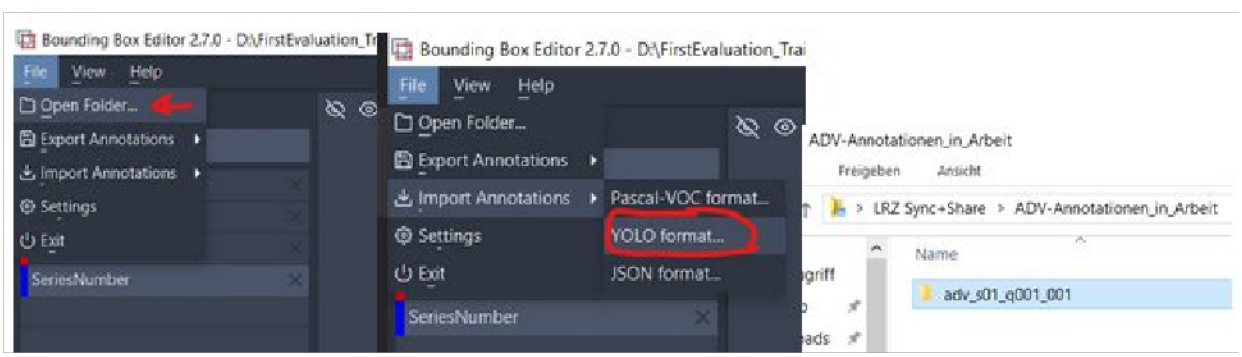

Figure 1: Workflow to start the annotation process

The user interface of the Bounding Box Editor is divided into three columns (Figure 1). The left column contains the relevant fields Categories and Objects, the middle column contains the workspace where the markings are set, and the right column contains thumbnails of the scans to be edited.

The categories in the left-hand column should be displayed automatically after the "Import Annotations" step. No other categories should be created. Similarly, categories that are already set up mustn't be deleted from the existing scheme.

In the "Categories" section, you can choose which category to mark (red arrow) (Figure 2). The "Objects" area clearly summarises the categories and the markers set for each category. If you hover the cursor over an entry in this area, the mark is highlighted in the main window and appears in a small preview window next to the cursor. Since it is very easy to accidentally set incorrect marks, especially at the beginning, this area is useful for checking the finished annotations. Incorrect markings can be deleted by right-clicking on the corresponding entry.

Another helpful feature is the ability to temporarily hide individual markings or entire markings. This is done using the small, coloured squares in the Objects area (red arrow). This function is useful when different markings overlap and would otherwise "block" each other (in Figure 2, for example, the red header marks an area that is also assigned to the yellow area Question).

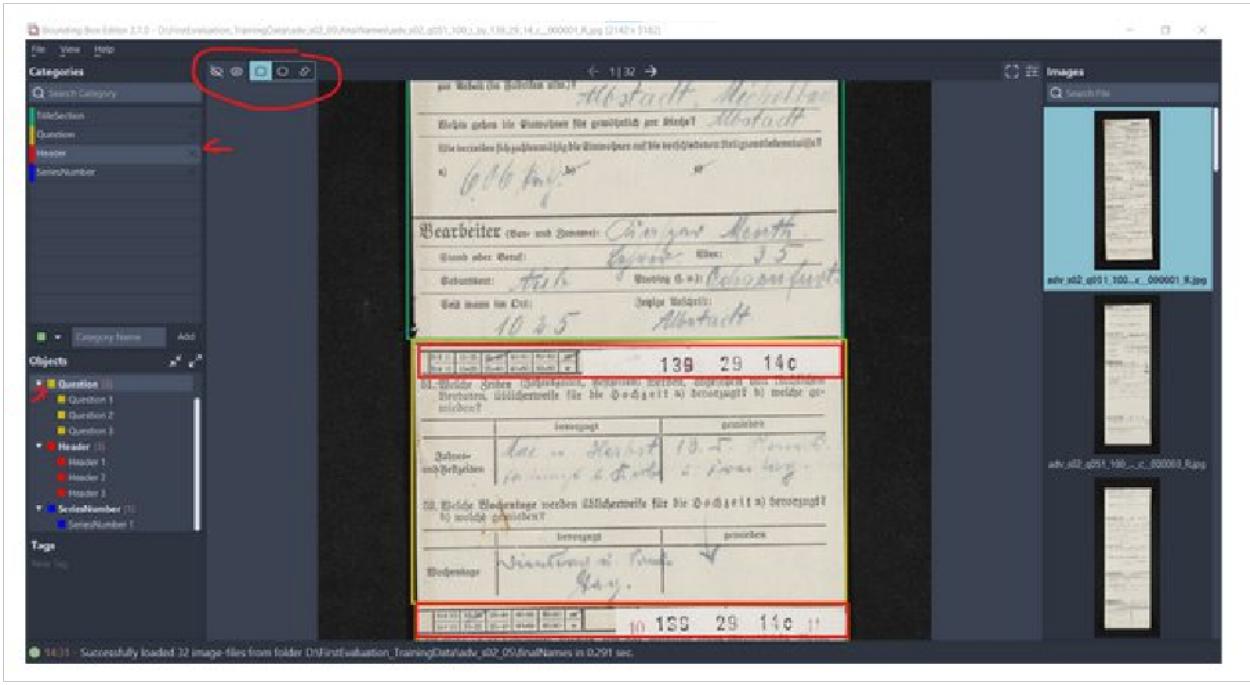

Figure 2: User Interface of the Bounding Box Editor

The markings are placed on the scans in the middle column. After the "Import Annotations" step, markings should already be in place on at least the first scan in the collection. These serve as sample examples for the following pages. During processing, all scanned pages are then marked accordingly.

A rectangular box is created by dragging it from one corner with the left mouse button pressed. The boxes can be moved and resized afterwards. A simple click with the left mouse button creates a box, which means it is very easy to accidentally create surplus boxes. These must be removed again. In principle, polygons can also be annotated using the Bounding Box Editor, which is why it is important to ensure that the rectangle selection is selected at the top left of the main window (circled in red). Once one page has been annotated, you can move on to the next pages using the arrows above the main window or the thumbnails.

Categories

The 19 layout categories of the layout annotation scheme can be subdivided in three main categories:
(1) Layout categories that mark information relating to the stamped location information within the header area of each question, being applied for administrative purposes by employees within the bureaucracy of the ADV project.
(2) Layout categories that are used to distinguish between handwritten and typewritten answer texts and highlight multi-line text instances.
(3) Layout categories that are used to annotate frequently occurring patterns of nontextual markings on behalf of the respondents.

The categories are the following:

- location_ID_part1
- location_ID_part2
- location_ID_part3
- location_ID_full
- location_region_ID
- answer_handwritten
- answer_typewritten
- answer_multiline_container
- mark_curly_bracket
- mark_drawing
- mark_quote

- mark_cross_out
- mark_underline
- mark_selection
- y/n_yes
- y/n_both
- y/n_none
- y/n_no

Location

We use five different layout categories to annotate the contents for the identification of the location. The information is a mix of numbers and letters and is normally located in the header region of the questionnaires and are typically written in typewriter.

The initial three-digit number is annotated as location_ID_part1. The second number is annotated as location_ID_part2. The third element of the location, a combined string of a number and one or two characters is annotated as location_ID_part3. Then all three parts of the location_ID should be annotated in the container category location_ID_full. The stamped capital letter acronym is annotated as location_region_ID.

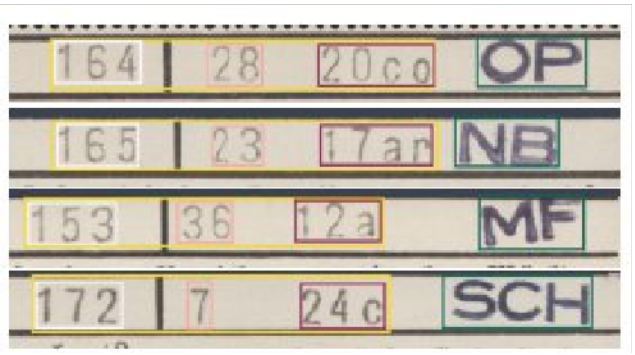

Figure 3: This figure shows examples for the location part of a questionnaire.

Figure 3 shows examples of the header. The components are labelled with separate bounding boxes. Location_ID_part1 as white, location_ID_part2 as orange, location_ID_part3 as red, the container category location_ID_full as yellow and location_region_ID as green.

Text answers

Written answers in the data occur in a multitude of different hand- and typewriting styles thus there are two different categories to annotate them. The separation is between handwritten answers with the category answer_handwritten and those printed by

typewriter with the category answer_typewritten. Only these two types should be distinguished. The font is ignored.

Because of the different kinds of the questionnaires the annotation of those answers can sometimes be confusing. As a rule, one line of the answer is annotated, and if an answer spans several lines, the container category answer_multiline_container should be used.

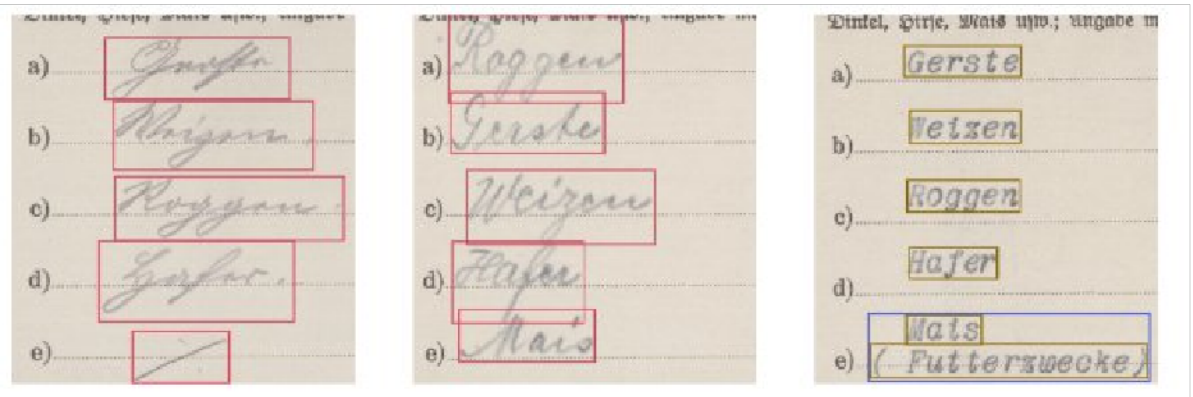

Figure 4: Examples for hand and typewritten answers

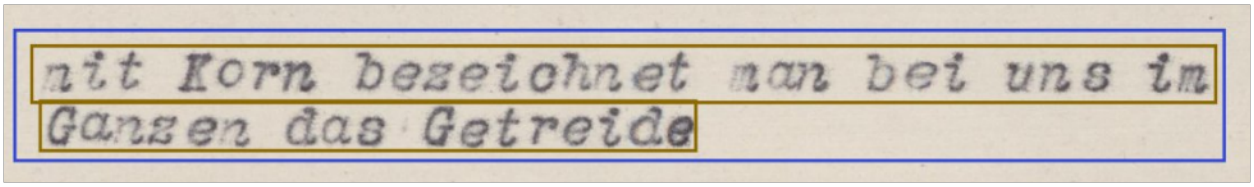

Figure 5: Example of a typewritten answer spanning two lines. The blue Bounding Box has the category answer_multiline_container

Often, if the question has not been answered, a placeholder is used to indicate this. For this there are the categories mark_omission or mark_quote. See chapter “non-textual marks for more information.

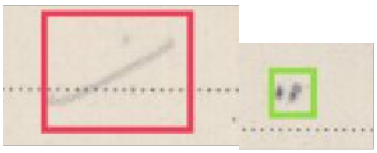

Figure 6: Examples off mark_omission (red) and mark_quote (green)

Sometimes handwritten text is typed over, in this case the handwritten text is only annotated, if the handwritten part can clearly be read and transcribed in later steps.

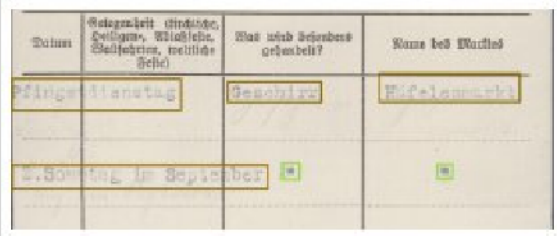

Figure 7: Here this is not the case. Only the typewritten answers are annotated.

Non-textual marks

To annotate the non-textual marks the following categories are to be used:

(1) mark underline: This label is assigned, whenever respondents underlined text of the pre-printed template. Mostly this action equals the positive selection of the underlined word. The Bounding Boxes should include the underscore and the underscored words.

(2) mark cross out: A crossing out of words operates in the opposite direction as a negative selection. As in the category above the bounding boxes include the underscore and the underscored or crossed out words. It is used in hand- and textwritten answers.

(3) mark omission: Bounding boxes with this label encompass lines across white space. The respondents wanted to signify that they would not or could not give a textual answer. Sometimes the omission line is accompanied by two dots, thus emulating a percentage sign (%).
It is also used when there is blank space crossed out i. e. there is no text beneath it.

(4) mark quote: Two short vertical lines – ditto marks – are often used to indicate that the preceding answer or line of text should be repeated at the location of the ”mark quote”.

(5) mark curly bracket: Curly brackets are often used in combination with a table template. Their intent is to assign an answer to multiple table rows. The respective bounding boxes only encompass the curly bracket itself.

(6) mark drawing: Bounding Boxes with this label encompass any figural drawings or sketches that were applied by a respondent.

(7) mark selection: In some instances respondents used little cross signs, for example to select one out of several pre-printed images.

Yes/No

The following selections are normally used for yes/no answers and can be a little bit tricky thus the following guidelines are very important.

The respective bounding box always encompasses the entire yes-no field. If nein (no) is crossed out, the bounding box is evaluated as yes-answer and labelled as y/n yes. The other options y/n no, y/n both and y/n none are used respectively, whenever ja (yes), none or both ja and nein are crossed out. The most common form of answering this type of questions is crossing out. However, if underlining is applied, the logic is to evaluate the underlined value as true. For example, if ja is underlined the bounding box around the yes-no field is labelled y/n yes.

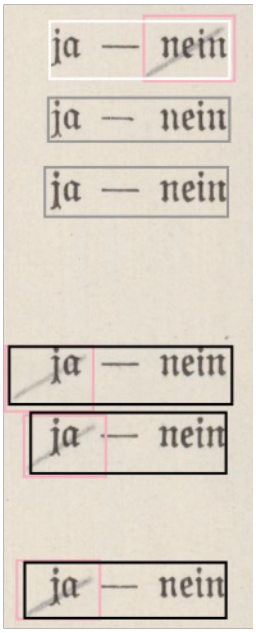

Figure 8: Examples for yes/no answers. In this case the categories are as following: y/n_yes (white), y/n_no (black), y/n_both (grey), mark_cross_out (pink).

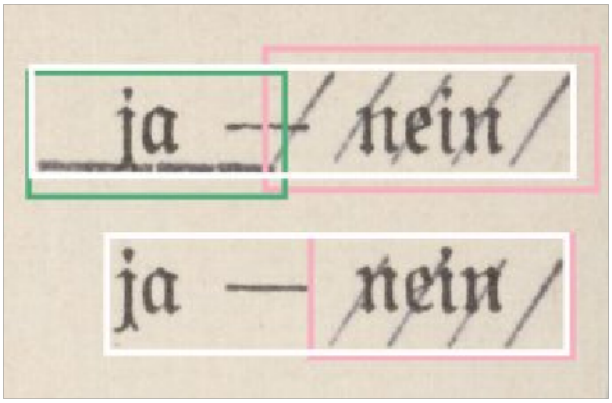

Figure 9: Another example for yes/no. Here there is in addition to mark_cross_out the category Mark_underline (green) to indicate the answer as yes.